

Case Study: Eliminating Rare-Earth Dependency with Advanced Self- Excited Synchronous Machines (SESM)

Client: Automotive OEMs & Industrial Propulsion Manufacturers

Prepared by: FEAAM GmbH

Subject: Implementing Advanced Brushless Architecture for Magnet-Free High-Torque Propulsion

Executive Summary

The reliance on Permanent Magnet Synchronous Machines (PMSM) has shifted from performance benefit to a strategic liability due to rare-earth price volatility and supply chain fragility. FEAAM GmbH has engineered an advanced Self-Excited Synchronous Machine (SESM) that provides a high-performance, "magnet-free" alternative. By utilizing a novel brushless architecture, this design delivers high torque capability even at zero speed—overcoming the primary limitation of previous brushless motors— without the need for expensive magnets or high-maintenance mechanical slip rings.

The Strategic Challenge: The Performance vs. Materials Gap

Manufacturers seeking to de-risk their supply chains have historically faced major trade-offs with magnet-free alternatives:

- Induction Motors (IM): Suffer from rotor losses that degrade efficiency and increase thermal complexity.
- Electrically Excited Motors (ESM): Rely on mechanical brushes and slip rings, which are prone to wear and limit power density.
- The "Zero-Speed" Barrier: Traditional brushless self-excited designs require motor motion to induce power. This leaves them unable to generate high torque at start-up (0 RPM).

The FEAAM Innovation: Advanced Brushless Architecture

FEAAM's SESM utilizes a proprietary structural design that enables contactless energy transfer to the rotor regardless of motor speed.

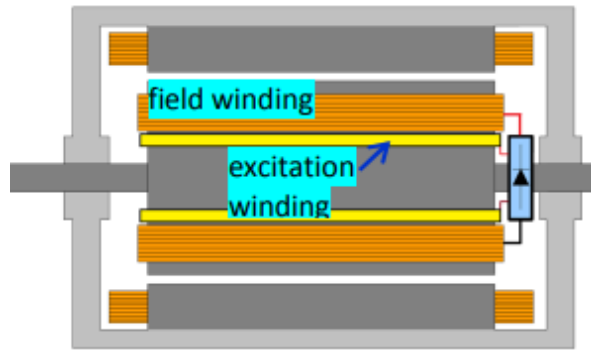


Fig. 1. Harmonic excitation.

Key Technical Breakthroughs:

- **Multiphase FSCW Stator:** Moves beyond traditional coil groups to wound each tooth individually, allowing independent control of magnetic waves.
- **Asymmetric Coil Design ($N_{w11} = N_{w12} + 1$):** A specialized configuration that boosts winding factors to increase power density and efficiency.
- **Advanced Rotor Topology:** Features two winding sets (Excitation and Field) with an integrated diode rectifier to capture energy directly from the stator's field.

Operational Excellence: Solving the Zero-Speed Torque Problem

The critical "unlock" is the multi-pole operation mode. By supplying the stator with a separate high-frequency excitation current, the motor induces voltage in the rotor based on the *excitation frequency* rather than *motor speed*.

- **Zero Speed:** High-frequency currents induce full torque capability immediately at 0 RPM.
- **High Speed:** At nominal and maximum speeds, the system transitions to using naturally occurring magnetic harmonics for excitation, maximizing energy economy.

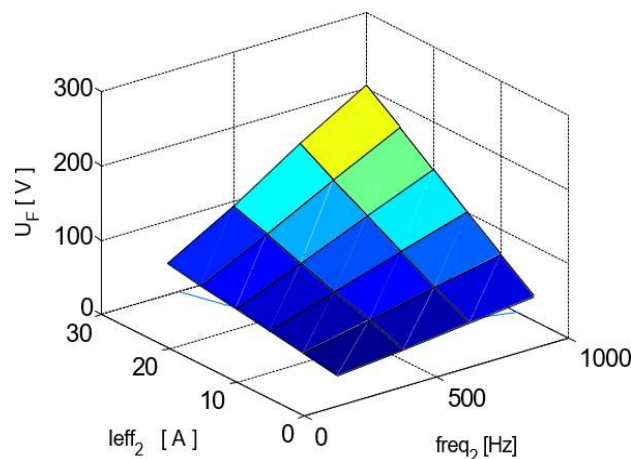


Fig. 2. Induced rotor DC voltage at rectifier terminals at zero rpm.

Performance Validation: Traction Benchmarking

FEAAM validated this architecture using a high-performance 11-phase, 8-pole motor model tailored for automotive traction.

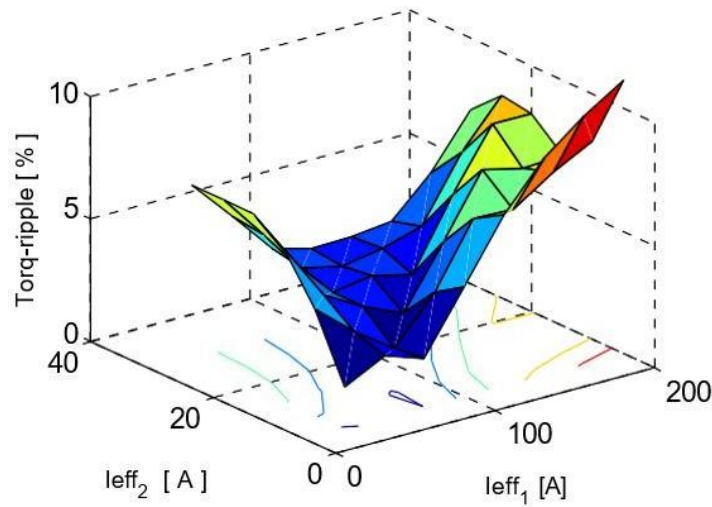


Fig. 3. Electromagnetic torque ripples at zero speed, and as function of load and excitation currents.

Performance Metric	Result	Strategic Benefit
Torque at Zero Speed	High Capability	Responsive, heavy load starts
Torque Ripple	< 2.7%	Premium smoothness and low noise
Rare-Earth Usage	0%	Total supply chain de-risking
Speed Range	0 – 14,000 RPM	High-performance highway capability

The Bottom Line for Business

The Advanced SESM allows OEMs to decouple their production success from the volatility of the rare-earth market.

- **Direct Cost Savings:** Eliminate the expense of Neodymium and Dysprosium magnets.
- **Zero Maintenance:** The brushless design removes high-wear components like slip rings, extending vehicle service life.
- **Future-Proof Performance:** Delivers the torque of a PMSM with the reliability of brushless architecture.

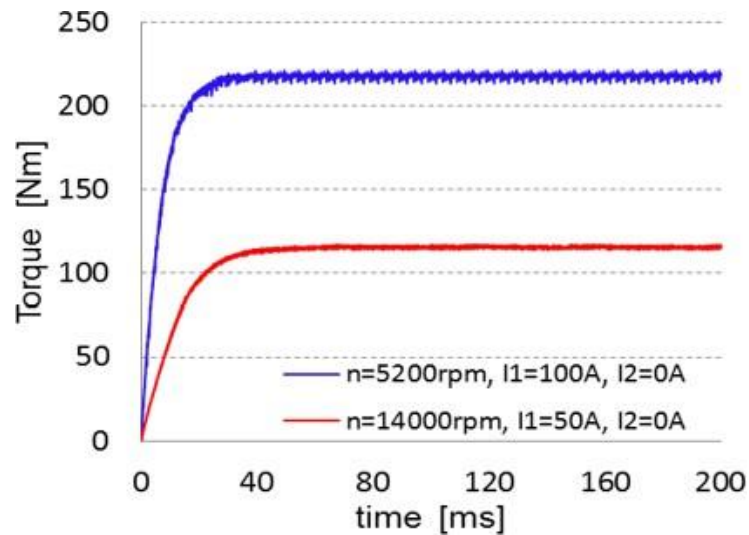


Fig. 4. Torque response for different operation conditions.

FEAAM can assist your team by integrating this proprietary winding and control topology into your existing platforms to deliver next-generation, sustainable propulsion.